MIL-W-0060307B(MU)
18 June 1973
USED IN LIEU OF
MIL-W-60307A
20 July 1971

MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET,

SMOKE, WP M156 (METAL PARTS)

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-W-60307A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers the metal parts for the M156 2.75 inch rocket warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum, and Aluminum Alloys.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 1340

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MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL90-1-39 - Warhead, 2.75 Inch Rocket, Smoke, WP, M156.

PICATINNY ARSENAL

F8837835 - Pallet for Projectile, Metal Parts.
C8861520 - Marking Diagram for Projectile, 75MM to 120MM.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions hould be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on DL90-1-39 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Assembly. All components shall be assembled in accordance with Drawing D90-1-39.

3.3 Leakage and distortion.

3.3.1 Warhead (hydrostatic). The warhead (Drawing D90-1-39) shall not leak or show evidence of distortion when subjected to an internal pressure of 5,000 ± 50 pounds per square inch gage (psig) for a period of 5 to 7 seconds at ambient temperature, when tested as specified in 4.4.4.1.

3.3.2 Warhead (air). The warhead accepted for hydrostatic leakage, shall not leak air when subjected to an internal air pressure of 100 ± 10 psig for a period of 15 to 20 seconds at ambient temperature when tested as specified in 4.4.4.2.

3.3.3 Burster tube casing. The burster tube casing shall not leak air when subjected to an internal pressure of 100 ± 10 psig for a period of 15 to 20 seconds at ambient temperature when tested as specified in 4.4.4.3.

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3.3.4 Brazed joint. When tested as specified in 4.4.4.4, the sectioned and peeled brazed joint shall be free of entrapped slag, unwetted or unbonded areas. The brazed joint shall be free of external defects as defined in MIL-B-7883. In addition the aggregate area of these defects in each joint shall not exceed 15 percent of the area of the faying surfaces. The maximum extent of a defect shall not exceed 15 percent of the overlap distance (width) of the joint. Presence of cleaning of other fluids in the solder groove shall be considered as evidence of a defective joint.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of warheads and burster tube casings shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The components shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks or deformation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herien, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth inthe specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 15 warheads and 15 burster tube casings shall be produced, using the same methods, materials, procedure and equipment as will be used during regular production.

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4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. The preproduction samples shall be examined and tested in accordance with the classification of defects. The AQL's do not apply.

4.3.2.2 For destructive test. Each brazed joint in the preproduction sample of warheads shall comply with 3.3.4 when tested as specified in 4.4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction samples shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168, and shall contain warhead components of one interfix number, produced by one manufacturer, in one unchanged process, at one plant, from the same materials in accordance with the same drawings, the same drawing revision, the same specification and the same specification revision.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. During the production of the first lot, one body assembly (Drawing D90-1-39) shall be randomly taken from each shift's production. After the first lot, six body assemblies shall be randomly taken from each succeeding lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample components shall be examined and tested in accordance with the classification of defects (4.4.3.3) and MIL-STD-105.

4.4.3.2 For destructive test. Prior to the application of protective coating and after air test (4.4.4.2), the brazed joints of the sample from the first lot shall be tested and examined in accordance with 4.4.4.4 for compliance with 3.3.4. In the event that a defective brazed joint is found, the contractor shall be required to provide written evidence to the

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Government of corrective action taken to preclude a recurrence of this defect prior to proceeding with production, and shall make a lot in accordance with the corrective action. Production shall not be initiated until test samples meet the requirements of 3.3.4.

4.4.3.3 Classification of defects.

(a) Adapter (Drawing C90-1-40).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Dimension 1.125 incorrect
102	Overall length incorrect
103	Surface finish incorrect (press fit area)
104	10° chamfer missing
Minor:	AQL 2.5 percent defective
201	Dimension 0.130 incorrect
202	Dimension 0.750 incorrect
203	Dimension 0.240 incorrect
204	Workmanship (see 3.5)

(b) Base (Drawing C90-1-41).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Dimension 1.92 incorrect
102	Dimension 2.23 incorrect

(c) Casing, Burster Tube (Drawing C90-1-28).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100 percent inspection	
101	Leakage	4.4.4.3
102	Inside diameter incorrect	

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4.4.3.3 Classification of defects (continued).

(c) Casing, Burster Tube (Drawing C90-1-28).

Categories	Defects	Acceptance standards
Major:	AQL 0.65 percent defective	
103	0.803 diameter and radius not continuous	
104	Taper incorrect	
105	0.400 dimension incorrect	
Major:	AQL 1.0 percent defective	
106	Surface finish incorrect (press fit area)	
107	7.475 dimension incorrect	
108	0.750 diameter incorrect	
109	0.652 dimension incorrect	
110	0.090 dimension (bottom thickness) incorrect	
111	0.8035 diameter incorrect	
112	Check concentricity of 0.8035 diameter to datum A	
113	Straightness of tube, inside contour and depth (note 4 on dwg)	
Minor:	AQL 2.5 percent defective	
201	0.06 radius incorrect	
202	0.75 dimension incorrect	
203	Workmanship (see 3.5)	
204	Wall thickness incorrect	

Note: All dimensions are applicable after painting.

(d) Warhead, 2.75 Inch Rocket, Smoke, WP, M156, (Metal Parts) (Drawing D90-1-39).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100 percent inspection	
101	Hydrostatic leakage	4.4.4.1
102	Air pressure leakage	4.4.4.2

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4.4.3.3 Classification of defects (continued).

(d) Warhead, 2.75 Inch Rocket, Smoke, WP, M156, (Metal Parts) (Drawing D90-1-39).

Categories	Defects	Acceptance Standards
Major:	AQL 0.40 percent defective	
103	Surface finish of 0.7985 diameter and 0.090 lead in and blend radius incorrect, surface analyzer with motor-trace	
104	Concentricity of 0.7985 diameter of surface D incorrect	
Major:	AQL 0.65 percent defective	
105	Diameter 0.7985 incorrect	
106	Diameter 2.225 incorrect	
Major:	AQL 1.0 percent defective	
107	Thread 1-7/16-16 UN-2B incorrect	
108	2.505 dimension incorrect	
109	10.395 dimension incorrect	
110	Concentricity 1.38 from nose incorrect	
111	Diameter 2.77 (surface A) incorrect	
112	Pitch or major diameter of rear thread incorrect	
113	Concentricity of 2.225 diameter to 2.77 diameter incorrect	
114	Diameter of bourrelet	
115	Concentricity of bourrelet diameter with outside diameter forward of bourrelet	
116	Concentricity of 2.225 diameter to surface F incorrect	
117	Point of tangency of 0.7985 diameter and the radius not concentric and without discontinuity - reverse taper (press fit surface)	
118	Perpendicularity of body to surface A incorrect	
119	Profile incorrect	

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4.4.3.3 Classification of defects (continued).

(d) Warhead, 2.75 Inch Rocket, Smoke, WP, M156, (Metal Parts) (Drawing D90-1-39)(continued).

Categories	Defects	Acceptance standards
Major:	AQL 1.0 percent defective	
120	Combination tolerance of concentricity to datum A and perpendicularity to surface C measured 0.25 and 2 inches from nose incorrect	
121	Workmanship (see 3.5)	

Note: All dimensions are applicable after painting.

Minor:	AQL 2.5 percent defective	
201	Diameter 2.275 incorrect	
202	Concentricity of diameter 1.455 to surface D incorrect	
203	Protective finish damaged	
204	Pitch diameter of rear thread maximum incorrect	

(e) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Strapping missing, loose or broken
102	Pallet damaged
103	Marking incorrect or illegible

Note: All dimensions are applicable after painting.

4.4.4 Tests.

4.4.4.1 Warhead (hydrostatic). The warhead shall be subjected to the specified pressure (using soluble oil and water mixture) for the required period of time. Accepted warheads shall be stamped with the letter "H",

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and the components shall be thoroughly cleaned and dried prior to the air test.

4.4.4.2 Warhead (air). After acceptance for hydrostatic leakage, and cleaned and dried per 4.4.4.1, the warhead shall be subjected to the specified pressure and completely submerged in water for the required time. Any indication of leakage shall be cause for rejection. Acceptable assembled components shall be stamped with the letter "A".

4.4.4.3 Burster tube casing. The burster tube casing shall be subjected to the specified pressure and completely submerged in water for the required period of time. Any indication of leakage shall be cause for rejection.

4.4.4.4 Brazed joint. The brazed joints on the sample warheads shall be sectioned and peeled from the warhead. The brazing and the solder groove on the warhead shall then be examined for compliance with 3.3.4.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and preservation. Warheads shall be clean and free of corrosion, and critical surfaces shall be preserved in accordance with applicable drawings and MIL-P-116.

5.2 Body packing, level C. The warheads shall be packed in accordance with Drawing F8837835.

5.3 Casing, burster tube, level C. The burster tube casing shall be packaged, packed, and marked in accordance with MIL-STD-1169.

5.4 Marking. Shipping containers shall be marked in accordance with Drawing C8861520.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The shipment group size and marking (when required).
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

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(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodians:

Army - MU

Preparing activity:

Army - MU(EA)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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